

Networking Basics

Lesson 40 — Networks, Topologies and Devices

JKSSB Computer Awareness

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What Is a Computer Network?

- ▶ A **computer network** is two or more computers (and other devices) connected so they can **share data and resources**.
- ▶ Shared resources include files, printers, an internet connection, and application software.
- ▶ Every participant in the network is called a **node**.
- ▶ A **host** is a node that can send or receive data over the network, such as a computer or a printer.

The one idea

A network connects nodes so that data and resources can be shared.

Client and Server

- ▶ A **client** is the device (or program) that **requests** a service, such as a web page or a file.
- ▶ A **server** is the computer that **provides** the service: it stores shared files or pages and answers client requests.
- ▶ In the common **client-server** arrangement, many clients ask and one server answers.
- ▶ The office desktop that opens a shared file is the client; the machine storing that file is the server.

Keep the pair straight

The client requests; the server provides. Do not swap the roles.

Network Types by Area: PAN, LAN, MAN, WAN

Type	Full form	Coverage
PAN	Personal Area Network	One person: phone, laptop, earphones
LAN	Local Area Network	One building or campus: office, school
MAN	Metropolitan Area Network	One city: branches linked across town
WAN	Wide Area Network	Country or world: the Internet is the largest WAN

Area grows from PAN (smallest) up to WAN (largest).

WLAN, Intranet and the Internet

- ▶ A **WLAN** (Wireless Local Area Network) is a LAN built with wireless signals instead of cables.
- ▶ An **intranet** is a private network inside one organisation; only its own staff can use it.
- ▶ The **Internet** is the worldwide public network of connected networks; anyone with access can use it.
- ▶ An **ISP** (Internet Service Provider) is the company that gives a customer access to the Internet.

Trap alert

An intranet is **private** to one organisation; the Internet is **public** and worldwide. Do not confuse the two words.

Network Topologies: What the Word Means

- ▶ A **topology** is the arrangement (layout) by which nodes are connected in a network.
- ▶ The five topologies in this lesson are **bus, star, ring, mesh, and tree**.
- ▶ Each topology decides how data travels and what happens when one cable or node fails.
- ▶ Exam questions ask: which topology has a central device? Which one fails fully when the main cable breaks?

How to remember them

Bus = one line; star = centre; ring = circle; mesh = every-to-every; tree = stars joined like branches.

Bus Topology

- ▶ In a **bus** topology, all nodes share **one single main cable** (the bus).
- ▶ It is simple and cheap for a small network.
- ▶ Data sent by one node travels along the whole bus and is received by the addressed node.
- ▶ If the **main cable breaks**, the whole network fails.

Bus: one-line weakness

One shared cable is the strength (cheap) and the weakness (one break stops everything).

Star Topology

- ▶ In a **star** topology, every node connects to one **central device** (a hub or switch).
- ▶ All data passes through the centre.
- ▶ If one cable fails, only that node is affected; the rest keep working.
- ▶ If the **central device** fails, the whole network stops.

Star is the office standard

The office LAN with one switch and many desktops is a star.

Ring Topology

- ▶ In a **ring** topology, each node connects to exactly **two neighbours**, forming a closed circle.
- ▶ Data travels around the ring, usually in one direction, from node to node until it reaches its destination.
- ▶ Each node passes on (repeats) the data it is not addressed to.
- ▶ A break at one point disturbs the whole ring unless a second (dual) ring provides a backup path.

Mesh and Tree Topologies

Mesh topology

- ▶ Every node has a direct link to many (or all) other nodes.
- ▶ Very **reliable**: if one path fails, another path carries the data.
- ▶ Costly: it needs the most cable and ports.

Tree topology

- ▶ Several **star** networks joined through a central backbone, like branches of a tree.
- ▶ Suits a large campus: each department is a star, joined at the centre.
- ▶ If the backbone fails, whole branches lose connection.

Mesh = most reliable and most costly; tree = stars joined together.

Topology Comparison at a Glance

Topology	Shape	Single point of failure
Bus	One shared cable	Main cable breaks: all fail
Star	Centre + spokes	Central hub/switch fails: all fail
Ring	Closed circle	One break disturbs the ring
Mesh	Many direct links	No single break stops it
Tree	Stars on a backbone	Backbone fails: branches fail

Read this table once and the matching items answer themselves.

Transmission Media: Guided and Unguided

- ▶ **Transmission media** are the paths that carry data signals from one device to another.
- ▶ **Guided (wired) media** confine the signal inside a cable: twisted-pair cable, coaxial cable, and fibre-optic cable.
- ▶ **Unguided (wireless) media** carry the signal through air or space: radio waves, microwaves, satellite links, and infrared.
- ▶ Guided media need a physical cable; unguided media need antennas or line of sight.

The one split

Guided = through a cable; unguided = through the air.

Guided Media: The Three Cables

- ▶ **Twisted-pair cable**: pairs of copper wires twisted together; the common telephone and office-LAN cable. Cheap, short range.
- ▶ **Coaxial cable**: a central copper wire with insulation and a shield; used for cable TV and older LANs. Better shielding than twisted pair.
- ▶ **Fibre-optic cable**: thin glass or plastic strands carrying **light** signals; fastest, longest range, immune to electrical noise, but costliest.

Fibre is fastest

Rank by speed and range: fibre-optic is fastest, then coaxial, then twisted pair is slowest of the three.

Unguided Media: Wireless Signals

- ▶ **Radio waves**: travel in all directions; used for FM radio and WLAN/Bluetooth over short ranges.
- ▶ **Microwaves**: travel in a straight line (line of sight); used for point-to-point links between towers.
- ▶ **Satellite links**: microwave signals relayed through a satellite; cover very long distances but with delay.
- ▶ **Infrared**: very short range, blocked by walls; used for TV remotes.

Networking Devices I: NIC, Modem, Hub

- ▶ **NIC** (Network Interface Card): the card (or built-in circuit) that lets a computer physically connect to a network.
- ▶ **Modem** (modulator-demodulator): converts digital computer signals to analogue signals for a telephone line and back again; it connects a home or office to the ISP.
- ▶ **Hub**: connects several nodes in a star and **broadcasts** every incoming signal to all ports; the addressed node accepts it.

Hub vs everything later

A hub sends data to **everyone**; a switch sends it only to the addressed device.

Networking Devices II: Switch, Bridge, Repeater

- ▶ **Switch**: connects nodes in a star and forwards each message **only to the addressed** port. Smarter and faster than a hub.
- ▶ **Bridge**: connects **two LAN segments** and filters traffic, passing only data meant for the other side.
- ▶ **Repeater**: **regenerates** a weak signal so it can travel further along a long cable; it works at the signal level only.

Trap alert

Hub = to all; switch = to the addressed port; repeater = only strengthens the signal, it does not choose a path.

Networking Devices III: Router, Gateway, Access Point, Firewall

- ▶ **Router**: connects **different networks** (for example a home LAN to the ISP) and **chooses the best path** for each packet.
- ▶ **Gateway**: connects networks that use **different protocols** or formats, translating between them.
- ▶ **Access point**: the wireless base station that lets Wi-Fi devices join a wired network.
- ▶ **Firewall**: **blocks unauthorised access** while letting legitimate traffic pass; a security device (hardware, software, or both).

Router vs gateway vs firewall

Router chooses the path; gateway translates between different networks; firewall guards the entry.

High-Yield Drill: Device–Role Matching

Device	One-line role
NIC	Connects a computer to the network
Modem	Digital ↔ analogue for the phone line / ISP link
Hub	Broadcasts to all ports
Switch	Forwards only to the addressed port
Repeater	Regenerates a weak signal
Bridge	Joins two LAN segments, filters traffic
Router	Joins different networks, chooses the path
Gateway	Translates between different protocols
Access point	Lets wireless devices join the network
Firewall	Blocks unauthorised access

Device–role matching is the highest-yield item type in this lesson.

Trap Alert: Networking Facts to Keep Straight

- ▶ A hub sends data to **all** nodes; a **switch** sends it only to the addressed node.
- ▶ A **repeater** only regenerates the signal; the **router** chooses the path.
- ▶ A **bridge** joins two similar LAN segments; a **gateway** joins (translates between) different protocols.
- ▶ **LAN** is one building; **MAN** is one city; **WAN** spans countries; **PAN** is one person.
- ▶ **Intranet** is private; the **Internet** is public.

Lesson 40: Recall

- ▶ A network links nodes to share data; a client requests and a server provides.
- ▶ By area: PAN (person), LAN (building), MAN (city), WAN (world); WLAN is wireless LAN; ISP = Internet Service Provider.
- ▶ Topologies: bus (one cable), star (centre), ring (circle), mesh (many links), tree (stars on a backbone).
- ▶ Guided media: twisted pair, coaxial, fibre-optic (fastest); unguided: radio, microwave, satellite, infrared.
- ▶ Devices: NIC connects; modem converts for the line; hub broadcasts; switch forwards to the address; repeater regenerates; bridge joins LAN segments; router paths between networks; gateway translates; access point admits Wi-Fi; firewall guards.